

Creative Scene

Objective: The goal of this assignment is for you to apply several of the computer graphics concepts we have learned in the course this semester.

Instructions:

- Using Three.js in CodePen, create a scene either individually or in a group that is either:
 - **Interactive** (responds to user input)
 - user interaction (mouse, keyboard, raycasting, UI controls)
 - **Animated** (moves or evolves over time)
 - animation using delta time (frame-independent motion)
- Your scene should utilize as many as possible, but **at least five** of the graphics concepts we have covered (in addition to being either interactive or animated):
 - Hierarchical modeling (parent-child relationships using `THREE.Group()`)
 - Transformations & coordinate systems (translation, rotation, scaling)
 - Relative positioning & spatial relationships (e.g., using bounding boxes, object alignment, collision bounds)
 - Texture mapping (applying image textures to objects (UV mapping))
 - Importing external 3D models (e.g., GLTF/GLB)
 - Environment mapping (e.g., creating a manual skybox (large cube/sphere with inward-facing textures), using `CubeTextureLoader` for environment maps)
 - Reflection, refraction, and Fresnel effects (e.g., using `MeshPhysicalMaterial` or dynamic environment maps with `CubeCamera`)
 - Particle systems / visual effects
 - Physics (e.g., using `Cannon.js`)
 - Custom shaders (GLSL)
- While it would be hard to do everything on this list, **please show a variety of skills**.
- If you do an animation, it should have at least **two** independently moving objects.

Grading will be in terms of the following requirements:

- (50%) Implementation of at least five of the topics named above
- (20%) Quality, cleanliness, and organization of code and documentation
- (15%) Creativity of the scene (not just redoing things we have already done throughout the semester)
- (10%) Interaction or Animation
- (5%) Completeness of description: clear and thorough reflection explaining the scene, the concepts used, and implementation details

Submission:

You will have two and a half weeks to complete this assignment.

When you have completed your assignment, **please submit the link to your CodePen project along with a brief reflection (3-4 paragraphs)** describing your submission and the features (see the list at the top of this page) that you demonstrated (maybe include a few screen shots of really cool things), along with any challenges you faced during implementation and how you overcame them.